Learning

How associations, rewards, punishments and role models shape behaviour — the psychology underneath every incentive scheme, loyalty programme and sales contest.

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If you have ever designed a dealer scheme, you have run a learning experiment. You chose a behaviour you wanted more of, attached a consequence, and decided when it would be paid. Psychology has studied exactly those three choices for over a century, with clear findings about which designs work and which backfire. This part covers the four kinds of learning — by association, by consequence, by watching others, and by thinking — and ends with a design checklist for incentives.

In psychology, **learning** means any relatively permanent change in behaviour brought about by experience. Changes from growing up or from fatigue don't count.

1 Classical conditioning: learning by association

In the 1890s the Russian physiologist Ivan Pavlov was studying digestion in dogs when he noticed they began to salivate before food arrived — at the sight of the assistant who fed them. He then paired a neutral sound with food. After repeated pairings, the sound alone produced salivation. The terms:

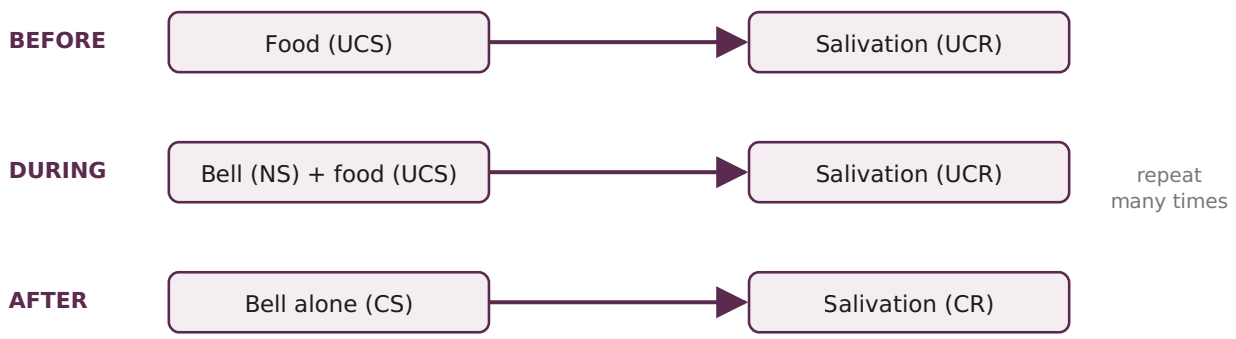


Figure 1. Classical conditioning. A neutral stimulus (NS) paired with an unconditioned stimulus (UCS) becomes a conditioned stimulus (CS) that produces the response on its own.

- **Unconditioned stimulus (UCS)** — triggers a response naturally, with no learning (food).
- **Unconditioned response (UCR)** — the natural response (salivation to food).
- **Conditioned stimulus (CS)** — a once-neutral stimulus that, after pairing, triggers the response (the bell).
- **Conditioned response (CR)** — the learned response to the CS (salivation to the bell).

Four further features matter in practice. **Extinction**: if the CS keeps appearing without the UCS, the response fades. **Spontaneous recovery**: after a break, an extinguished response briefly returns. **Generalisation**: similar stimuli trigger the response too. **Discrimination**: with training, the learner responds only to the exact stimulus.

Where it shows up in business

Advertising is largely classical conditioning: pair a brand (neutral) with music, celebrities, happiness or festivals (stimuli that already produce good feelings), often enough that the brand alone produces a warm response. Generalisation explains why look-alike packaging works for copycat brands, and why a scandal around one product can hurt the whole range. Conditioning also works against you. A dealer who is shouted at during every visit learns to feel tense when your rep's bike pulls up.

Two classic studies set the limits. In 1920 John Watson and Rosalie Rayner conditioned an infant, "Little Albert", to fear a white rat by pairing it with a loud noise; the fear generalised to other furry things. The study would be unethical today and was methodologically weak, but it showed emotions can be conditioned. And in the 1960s John Garcia found that rats learned to avoid a taste after a *single* pairing with nausea hours later — **conditioned taste aversion**. Not all associations are equally easy: we are biologically prepared to learn some, like linking food with illness, very fast. One bad experience with a product can do what a hundred good adverts can't undo.

2 Operant conditioning: learning by consequences

Classical conditioning is about involuntary responses. **Operant conditioning** is about voluntary behaviour and what follows it. Edward Thorndike's **law of effect** (1898): behaviour followed by a pleasing consequence is more likely to be repeated; behaviour followed by an unpleasant one, less likely. B. F. Skinner turned this into a precise science from the 1930s.

The vocabulary trips everyone up, so note it carefully. **Reinforcement** always *increases* a behaviour; **punishment** always *decreases* it. **Positive** means something is added; **negative** means something is taken away.

	Add something	Remove something
Increase behaviour	**Positive reinforcement** — a bonus for hitting target	**Negative reinforcement** — no more daily call-reports once a rep's pipeline is healthy
Decrease behaviour	**Positive punishment** — a public reprimand for a missed target	**Negative punishment** — losing the company car for a quarter

"Negative reinforcement" is not punishment. It strengthens behaviour by removing something unpleasant — and it is one of the most underused tools in management. Freedom from micromanagement is a powerful reward.

Why punishment usually disappoints

Punishment can stop a behaviour quickly, but it has known side effects. It tells people what not to do, not what to do. It works mainly when the punisher is present, so the behaviour goes underground. It produces fear and resentment toward the person punishing. And it models aggression. Punishment works best when immediate, consistent, and paired with reinforcement of the right alternative behaviour.

Shaping

Complex behaviour rarely appears fully formed to be rewarded. **Shaping** reinforces successive approximations: reward the first step toward the goal, then require a bit more before the next reward. A new rep is praised for completing a dealer call, then for a call with a proper order review, then for a call that brings in a new product line.

3 Schedules of reinforcement

When the reward arrives matters as much as what it is. **Continuous reinforcement** (every time) teaches a new behaviour fastest but extinguishes fastest when rewards stop. **Partial reinforcement** is slower to learn but far more durable. There are four partial schedules:

Schedule	Rule	Typical behaviour	Example
Fixed ratio (FR)	Reward after every *n* responses	High rate, pause after each reward	Commission on every 10 units
Variable ratio (VR)	Reward after an unpredictable number	Highest, steadiest rate; hardest to extinguish	Lottery, scratch cards, cold calling
Fixed interval (FI)	First response after a set time is rewarded	Slow start, rush just before the deadline ("scallop")	Quarterly target bonus
Variable interval (VI)	First response after an unpredictable time	Slow but steady	Surprise audits, random checks

WORKED EXAMPLE 1 · THE QUARTER-END HOCKEY STICK

A company pays its sales team on a quarterly target — a fixed-interval schedule. A typical territory's quarter (illustrative):

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Month 1:  25% of quarterly volume
Month 2:  30%
Month 3:  45%   ← the "scallop"

Month 3 vs month 1: 45 / 25 = 1.8 times
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This is exactly what fixed-interval schedules produce in the lab: little effort early, a burst near the deadline. In channel sales the burst is often "loading" — pushing stock onto distributors with extra discounts — which then depresses the next quarter's first month. Remedies follow from the schedule: monthly or weekly targets (shorter intervals), rewards tied to secondary sales (what dealers actually sell on), or some variable elements so effort pays off throughout the quarter.

WORKED EXAMPLE 2 · PRICING A VARIABLE-RATIO SCHEME

A painter loyalty programme gives one scratch card per 10-litre purchase. Prizes: 1 card in 20 wins ₹1,000; 1 card in 5 wins ₹50; the rest win nothing.

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Expected value per card
  = (1/20 × ₹1,000) + (1/5 × ₹50)
  = ₹50 + ₹10 = ₹60
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It costs the same on average as a flat ₹60 cashback per purchase. But a variable-ratio schedule tends to produce steadier engagement and is slower to extinguish once prizes stop, and the occasional ₹1,000 winner becomes word-of-mouth. Two caveats. Payouts must be real and visible, or trust collapses. And this is the same mechanism that makes gambling compelling, so it should never be designed to exploit people who can't afford it.

4 Learning by watching

We don't have to be rewarded ourselves to learn. In Albert Bandura's 1961 **Bobo doll** studies, children who watched an adult attack an inflatable doll later copied the attacks, often with the same words; children who watched a calm adult did not. **Observational learning** needs four things: **attention** to the model, **memory** of what they did, the **ability** to reproduce it, and **motivation** to do so — often from seeing the model rewarded (**vicarious reinforcement**).

The management implication is direct: teams copy what leaders do and what gets rewarded, far more than what the policy says. If the rep who quietly over-promises to dealers wins "Salesman of the Quarter", everyone has just learned the real rule. Role models, ride-alongs and shadowing are observational learning on purpose.

5 Learning by thinking

- **Latent learning** — Edward Tolman found rats that explored a maze without reward learned its layout anyway, and used it immediately once food appeared. Learning can happen without showing in behaviour until there's a reason to use it. A new rep riding along for a month may know more than his reports show.
- **Insight** — Wolfgang Köhler's chimpanzees, after failing to reach bananas, suddenly stacked boxes or joined sticks. Some problems are solved by a sudden restructuring, not by trial and error.
- **Learned helplessness** — Martin Seligman found that animals exposed to shocks they couldn't control later stopped trying to escape even when escape was possible. In 2016 Seligman and Steven Maier revised the explanation: passivity is the brain's default response to prolonged bad events, and what has to be learned is that you have control. The workplace version is a rep who keeps missing targets set impossibly high, and who stops trying. Restoring some control — targets he can influence, small wins — is the fix.

6 Held up?

Claim	Verdict	Why
Classical and operant conditioning, schedule effects	**Robust**	Among the most replicated findings in psychology, across species.
Rewards kill intrinsic motivation ("overjustification")	**Conditional**	Expected, tangible rewards for an activity people already enjoy can reduce their interest once rewards stop. For dull tasks, or for unexpected and verbal rewards, the effect is small or absent. Still debated.
"10,000 hours of practice makes an expert"	**Overstated**	A 2014 meta-analysis found deliberate practice explained about a quarter of performance differences in games, a fifth in music, about a sixth in sports — and under 1% in professions. Practice matters; it isn't the whole story.
Little Albert proves phobias are conditioned	**Weak evidence**	One child, poor controls. Conditioning does contribute to fears, from better later studies.
Children imitate aggression they watch	**Holds**	Repeated in lab settings; how much it explains real-world violence is debated.

INCENTIVE DESIGN CHECKLIST

1. Reward the behaviour you want, not a proxy that can be gamed (secondary sales, not loading).
2. Keep the gap between behaviour and reward short.
3. Choose the schedule deliberately; long fixed intervals create deadline rushes.
4. Use negative reinforcement: remove hassle as a reward.
5. Punish sparingly, immediately and with a clear alternative.
6. Watch what you publicly reward — everyone else is learning from it.

Summary

- **Learning** is a relatively permanent change in behaviour due to experience.
- **Classical conditioning** pairs a neutral stimulus with one that naturally triggers a response, until the neutral one triggers it alone. It underlies brand advertising; extinction, generalisation and discrimination follow predictable rules. Some associations, like taste aversion, form in one trial.
- **Operant conditioning**: consequences shape voluntary behaviour. Reinforcement increases behaviour, punishment decreases it; positive adds, negative removes. Punishment has side effects; shaping builds complex behaviour step by step.
- **Schedules**: variable ratio gives the highest, most durable response; fixed interval produces deadline rushes — the quarter-end hockey stick.
- **Observational learning** (Bandura) needs attention, memory, ability and motivation. Teams copy what leaders model and reward.
- **Cognitive learning**: latent learning, insight, and learned helplessness, now understood as the default that a sense of control reverses.
- Rewards can undermine interest in enjoyable tasks under some conditions; "10,000 hours" overstates the role of practice.

Practice problems

P1. A paint brand plays the same jingle in every Diwali ad for five years. Now the jingle alone makes viewers feel festive. Identify the UCS, UCR, CS and CR.

P2. Classify each: (a) a rep is excused from the Monday 8 am call after hitting target; (b) a dealer loses his credit line after late payment; (c) a rep gets a trip to Goa for top sales; (d) a manager criticises a rep in front of the team.

P3. A scheme gives a contractor a ₹500 voucher for every 50 litres purchased. Which schedule is this? Predict the behaviour pattern.

P4. A scratch card offers 1-in-50 odds of ₹2,000 and 1-in-10 odds of ₹100. What is its expected value?

P5. A territory's monthly split across a quarter is 20%, 30%, 50%. What's driving this, and suggest two design changes.

P6. A rep who missed target three quarters running has stopped making new-dealer calls. Using this part, explain why and what you would change.

Glossary

Learning

A relatively permanent change in behaviour due to experience.

UCS / UCR

A stimulus that naturally triggers a response / that natural response.

CS / CR

A learned trigger / the learned response to it.

Extinction

Fading of a learned response when it is no longer paired or reinforced.

Generalisation / discrimination

Responding to similar stimuli / only to the specific one.

Law of effect

Behaviour followed by pleasing consequences is repeated.

Reinforcement / punishment

A consequence that increases / decreases a behaviour.

Positive / negative

Something is added / removed.

Shaping

Reinforcing successive steps toward a target behaviour.

Partial reinforcement

Rewarding only some responses; slower to learn but more durable.

Fixed / variable ratio

Reward after a set / an unpredictable number of responses.

Fixed / variable interval

Reward for the first response after a set / an unpredictable time.

Observational learning

Learning by watching others.

Vicarious reinforcement

Being motivated by seeing someone else rewarded.

Latent learning

Learning that doesn't show until there's a reason to use it.

Learned helplessness

Giving up after experiences of having no control.

Overjustification effect

External rewards reducing interest in a task people already enjoyed.

Answer key

P1. UCS: the festive imagery and family moments. UCR: the warm, festive feeling. CS: the jingle. CR: the festive feeling triggered by the jingle alone.

P2. (a) negative reinforcement; (b) negative punishment; (c) positive reinforcement; (d) positive punishment.

P3. Fixed ratio (every 50 litres). Expect a high purchase rate, with a pause after each voucher, and possibly orders bunched into 50-litre blocks.

P4. (1/50 × ₹2,000) + (1/10 × ₹100) = ₹40 + ₹10 = **₹60**.

P5. A fixed-interval quarterly reward creates a deadline scallop. Fixes: monthly targets or milestones; reward secondary sales; add a variable element such as surprise spot prizes for weekly activity.

P6. Repeated failure he couldn't control has produced learned helplessness: passivity is the default once effort seems pointless. Restore control — a target he can influence, small, quick wins, and reinforcement for the activity (new-dealer calls), not just the result.

Sources: chapter structure follows Ciccarelli & White, *Psychology*, 4th ed. (Pearson), Ch. 5; text and figures are original. Bobo doll: Bandura, Ross & Ross, 1961. Learned helplessness revision: Maier & Seligman, *Psychological Review*, 2016. Overjustification: Deci, Koestner & Ryan, *Psychological Bulletin*, 1999. Practice: Macnamara, Hambrick & Oswald, *Psychological Science*, 2014. Worked-example figures are illustrative.